Tang Nano 20K

Revision History

Version	Date	Change Note
3850	2022/12/24	Original edition
3920	2023/02/22	Shorten PCB width to 888mil Changed pin assignment of BL616.
3921	2023/03/27	Fix some Silkscreen issuses First public release version.

GW2AR-18

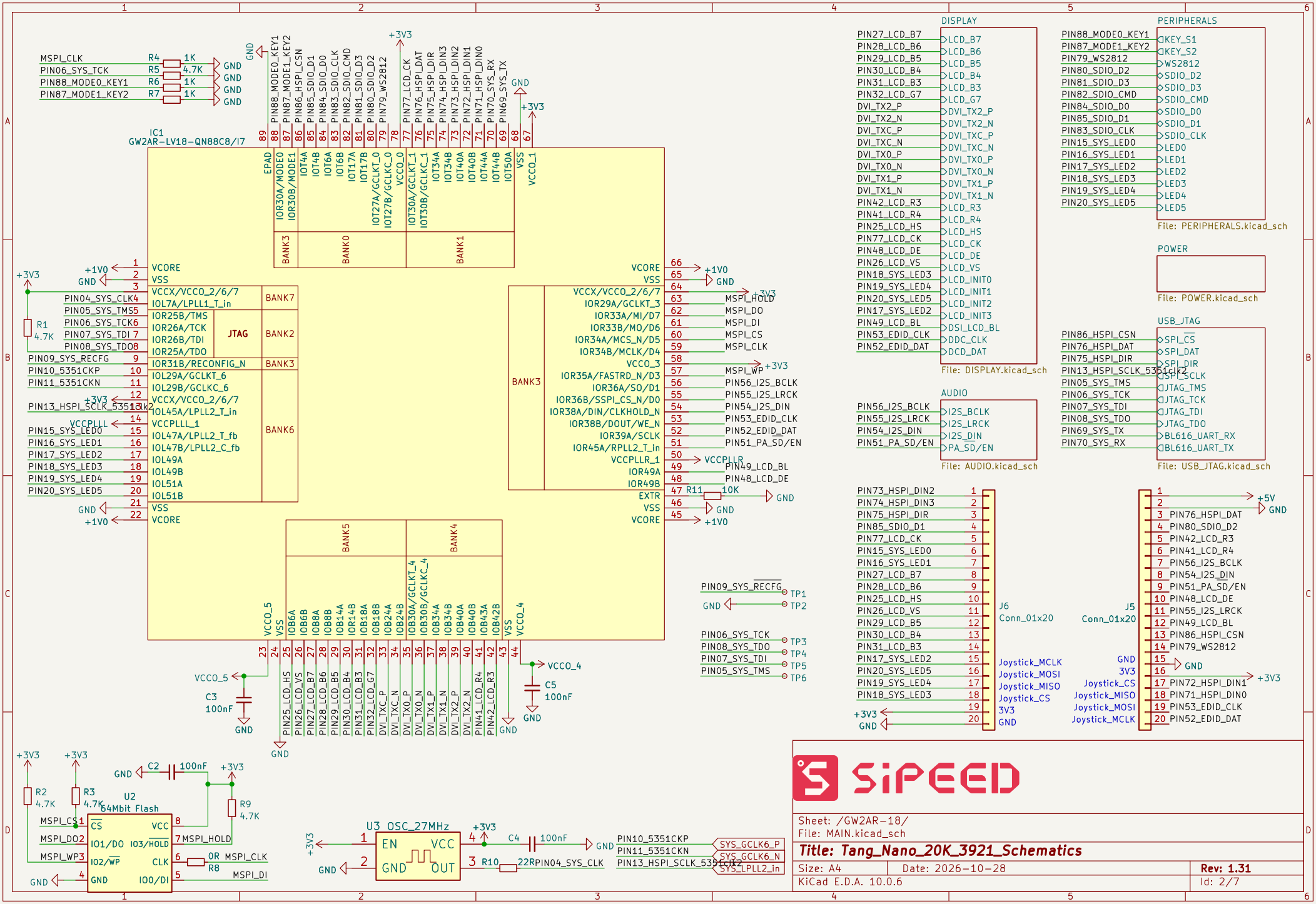
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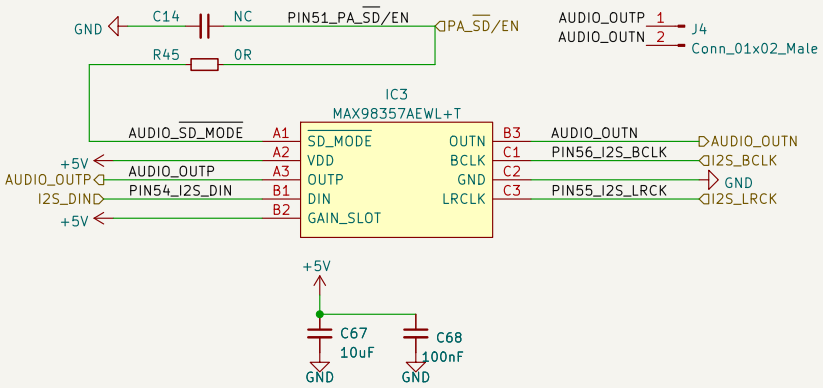
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Title: Tang_Nano_20K_3921_Schematics

Size: A4	Date: 2026-10-28	**Rev: 1.31**
KiCad E.D.A. 10.0.6		Id: 1/7


IC1
GW2AR-LV18-QN88C8/I7
MSPI_CLK
PIN06_SYS_TCK
PIN88_MODE0_KEY1
PIN87_MODE1_KEY2
R4 1K
R5 4.7K
R6 1K
R7 1K
GND
+3V3
PIN88_MODE0_KEY1
PIN87_MODE1_KEY2
PIN86_HSPI_CSN
PIN85_SDIO_D1
PIN84_SDIO_D0
PIN83_SDIO_CLK
PIN82_SDIO_CMD
PIN81_SDIO_D3
PIN80_SDIO_D2
PIN79_WS2812
PIN77_LCD_CK
PIN76_HSPI_DAT
PIN75_HSPI_DIR
PIN74_HSPI_DIN3
PIN73_HSPI_DIN2
PIN72_HSPI_DIN1
PIN71_HSPI_DIN0
PIN70_SYS_RX
PIN69_SYS_TX
EPAD
IOR30A/MODE0
IOR30B/MODE1
IOT4A
IOT4B
IOT6A
IOT6B
IOT17A
IOT17B
IOT27A/GCLKT_0
IOT27B/GCLKC_0
VCCO_0
IOT30A/GCLKT_1
IOT30B/GCLKC_1
IOT34A
IOT34B
IOT40A
IOT40B
IOT44A
IOT44B
IOT50A
VSS
VCCO_1
BANK3
BANK0
BANK1
VCORE
VSS
VCCX/VCCO_2/6/7
IOL7A/LPLL1_T_in
IOR25B/TMS
IOR26A/TCK
IOR26B/TDI
IOR25A/TDO
JTAG
IOR31B/RECONFIG_N
IOL29A/GCLKT_6
IOL29B/GCLKC_6
IOL45A/LPLL2_T_in
VCCPLLL_1
IOL47A/LPLL2_T_fb
IOL47B/LPLL2_C_fb
IOL49A
IOL49B
IOL51A
IOL51B
BANK7
BANK2
BANK3
BANK6
PIN04_SYS_CLK
PIN05_SYS_TMS
PIN06_SYS_TCK
PIN07_SYS_TDI
PIN08_SYS_TDO
PIN09_SYS_RECFG
PIN10_5351CKP
PIN11_5351CKN
PIN13_HSPI_SCLK_5351CLK2
PIN15_SYS_LED0
PIN16_SYS_LED1
PIN17_SYS_LED2
PIN18_SYS_LED3
PIN19_SYS_LED4
PIN20_SYS_LED5
R1 4.7K
+1V0
VCCPLLL
VCCX/VCCO_2/6/7
IOR29A/GCLKT_3
IOR33A/MI/D7
IOR33B/MO/D6
IOR34A/MCS_N/D5
IOR34B/MCLK/D4
VCCO_3
IOR35A/FASTRD_N/D3
IOR36A/SO/D1
IOR36B/SSPI_CS_N/D0
IOR38A/DIN/CLKHOLD_N
IOR38B/DOUT/WE_N
IOR39A/SCLK
IOR45A/RPLL2_T_in
VCCPLLR_1
IOR49A
IOR49B
EXTR
BANK3
MSPI_HOLD
MSPI_DO
MSPI_DI
MSPI_CS
MSPI_CLK
MSPI_WP
PIN56_I2S_BCLK
PIN55_I2S_LRCK
PIN54_I2S_DIN
PIN53_EDID_CLK
PIN52_EDID_DAT
PIN51_PA_SD/EN
VCCPLLR
PIN49_LCD_BL
PIN48_LCD_DE
R11 10K
BANK5
BANK4
VCCO_5
VCCO_4
IOB6A
IOB6B
IOB8A
IOB8B
IOB14A
IOR14B
IOB18A
IOB18B
IOB24A
IOB24B
IOB30A/GCLKT_4
IOB30B/GCLKC_4
IOB34A
IOB34B
IOB40A
IOB40B
IOB43A
IOB42B
PIN25_LCD_HS
PIN26_LCD_VS
PIN27_LCD_B7
PIN28_LCD_B6
PIN29_LCD_B5
PIN30_LCD_B4
PIN31_LCD_B3
PIN32_LCD_G7
DVI_TXC_P
DVI_TXC_N
DVI_TX0_P
DVI_TX0_N
DVI_TX1_P
DVI_TX1_N
DVI_TX2_P
DVI_TX2_N
PIN41_LCD_R4
PIN42_LCD_R3
C3 100nF
C5 100nF
PIN09_SYS_RECFG
PIN06_SYS_TCK
PIN08_SYS_TDO
PIN07_SYS_TDI
PIN05_SYS_TMS
TP1
TP2
TP3
TP4
TP5
TP6
DISPLAY
PIN27_LCD_B7
PIN28_LCD_B6
PIN29_LCD_B5
PIN30_LCD_B4
PIN31_LCD_B3
PIN32_LCD_G7
DVI_TX2_P
DVI_TX2_N
DVI_TXC_P
DVI_TXC_N
DVI_TX0_P
DVI_TX0_N
DVI_TX1_P
DVI_TX1_N
PIN42_LCD_R3
PIN41_LCD_R4
PIN25_LCD_HS
PIN77_LCD_CK
PIN48_LCD_DE
PIN26_LCD_VS
PIN18_SYS_LED3
PIN19_SYS_LED4
PIN20_SYS_LED5
PIN17_SYS_LED2
PIN49_LCD_BL
PIN53_EDID_CLK
PIN52_EDID_DAT
LCD_B7
LCD_B6
LCD_B5
LCD_B4
LCD_B3
LCD_G7
LCD_R3
LCD_R4
LCD_HS
LCD_CK
LCD_DE
LCD_VS
LCD_INIT0
LCD_INIT1
LCD_INIT2
LCD_INIT3
DSI_LCD_BL
DDC_CLK
DCD_DAT
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PERIPHERALS
PIN88_MODE0_KEY1
PIN87_MODE1_KEY2
PIN79_WS2812
PIN80_SDIO_D2
PIN81_SDIO_D3
PIN82_SDIO_CMD
PIN84_SDIO_D0
PIN85_SDIO_D1
PIN83_SDIO_CLK
PIN15_SYS_LED0
PIN16_SYS_LED1
PIN17_SYS_LED2
PIN18_SYS_LED3
PIN19_SYS_LED4
PIN20_SYS_LED5
KEY_S1
KEY_S2
WS2812
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_D0
SDIO_D1
SDIO_CLK
LED0
LED1
LED2
LED3
LED4
LED5
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POWER
File: POWER.kicad_sch
AUDIO
PIN56_I2S_BCLK
PIN55_I2S_LRCK
PIN54_I2S_DIN
PIN51_PA_SD/EN
I2S_BCLK
I2S_LRCK
I2S_DIN
PA_SD/EN
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USB_JTAG
PIN86_HSPI_CSN
PIN76_HSPI_DAT
PIN75_HSPI_DIR
PIN13_HSPI_SCLK_5351CLK2
PIN05_SYS_TMS
PIN06_SYS_TCK
PIN07_SYS_TDI
PIN08_SYS_TDO
PIN69_SYS_TX
PIN70_SYS_RX
SPI_CS
SPI_DAT
SPI_DIR
SPI_SCLK
JTAG_TMS
JTAG_TCK
JTAG_TDI
JTAG_TDO
BL616_UART_RX
BL616_UART_TX
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J6
Conn_01x20
PIN73_HSPI_DIN2
PIN74_HSPI_DIN3
PIN75_HSPI_DIR
PIN85_SDIO_D1
PIN77_LCD_CK
PIN15_SYS_LED0
PIN16_SYS_LED1
PIN27_LCD_B7
PIN28_LCD_B6
PIN25_LCD_HS
PIN26_LCD_VS
PIN29_LCD_B5
PIN30_LCD_B4
PIN31_LCD_B3
PIN17_SYS_LED2
PIN20_SYS_LED5
PIN19_SYS_LED4
PIN18_SYS_LED3
Joystick_MCLK
Joystick_MOSI
Joystick_MISO
Joystick_CS
3V3
GND
J5
Conn_01x20
+5V
PIN76_HSPI_DAT
PIN80_SDIO_D2
PIN42_LCD_R3
PIN41_LCD_R4
PIN56_I2S_BCLK
PIN54_I2S_DIN
PIN51_PA_SD/EN
PIN48_LCD_DE
PIN55_I2S_LRCK
PIN49_LCD_BL
PIN86_HSPI_CSN
PIN79_WS2812
PIN72_HSPI_DIN1
PIN71_HSPI_DIN0
PIN53_EDID_CLK
PIN52_EDID_DAT
R2 4.7K
R3 4.7K
U2
64Mbit Flash
MSPI_CS
MSPI_DO
MSPI_WP
MSPI_HOLD
MSPI_CLK
MSPI_DI
C2 100nF
R9 4.7K
R8 0R
U3 OSC_27MHz
C4 100nF
R10 22R
PIN10_5351CKP
PIN11_5351CKN
PIN13_HSPI_SCLK_5351CLK2
SYS_GCLK6_P
SYS_GCLK6_N
SYS_PLL2_in
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Title: Tang_Nano_20K_3921_Schematics
Size: A4
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DAC + Audio PA



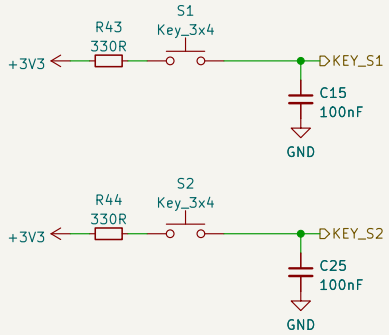
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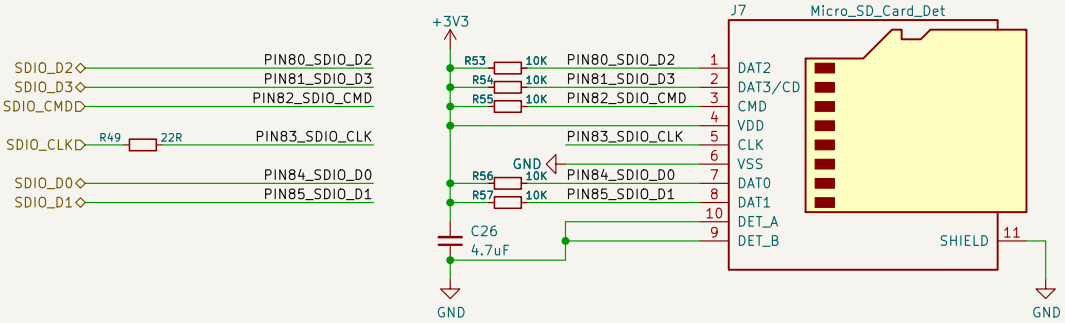
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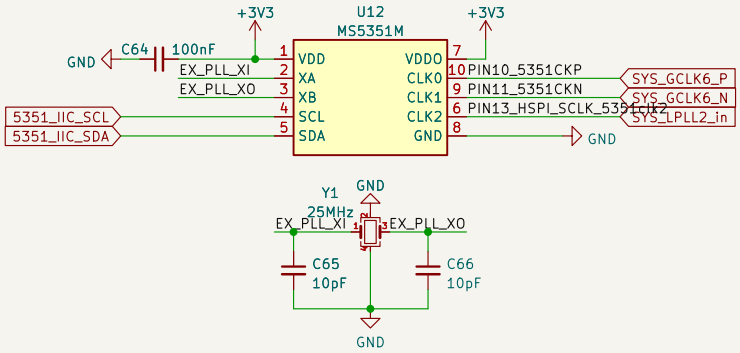
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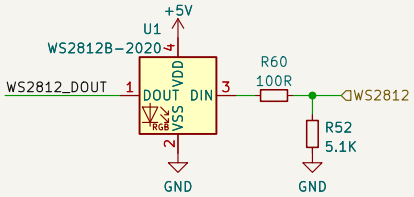
MICRO SD CARD



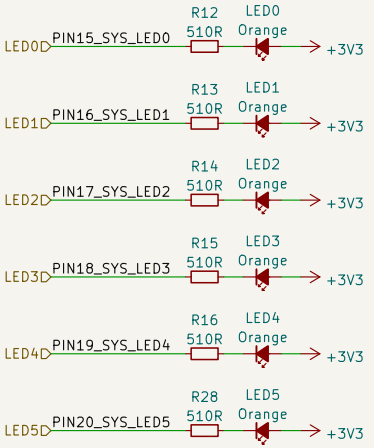
AUX_PLL



WS2812



LED

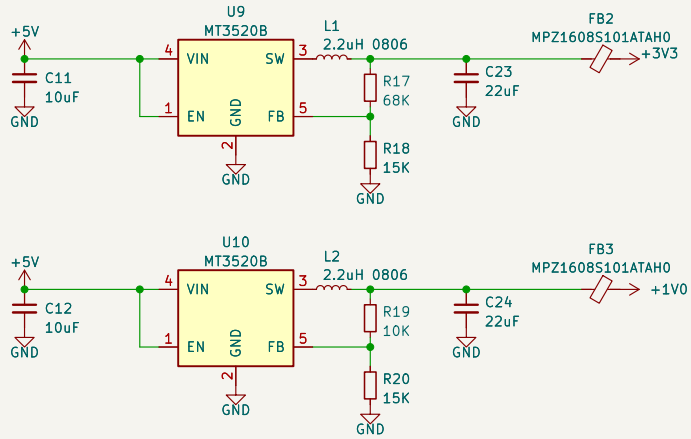


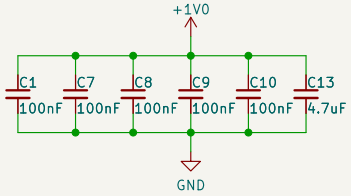
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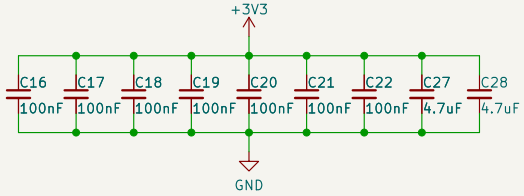
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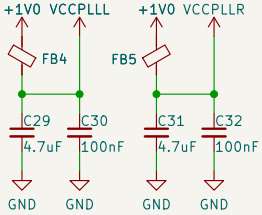
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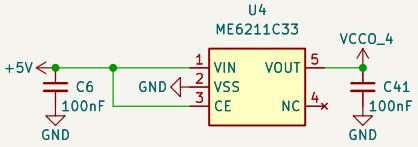
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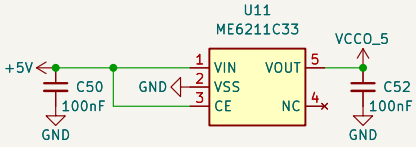










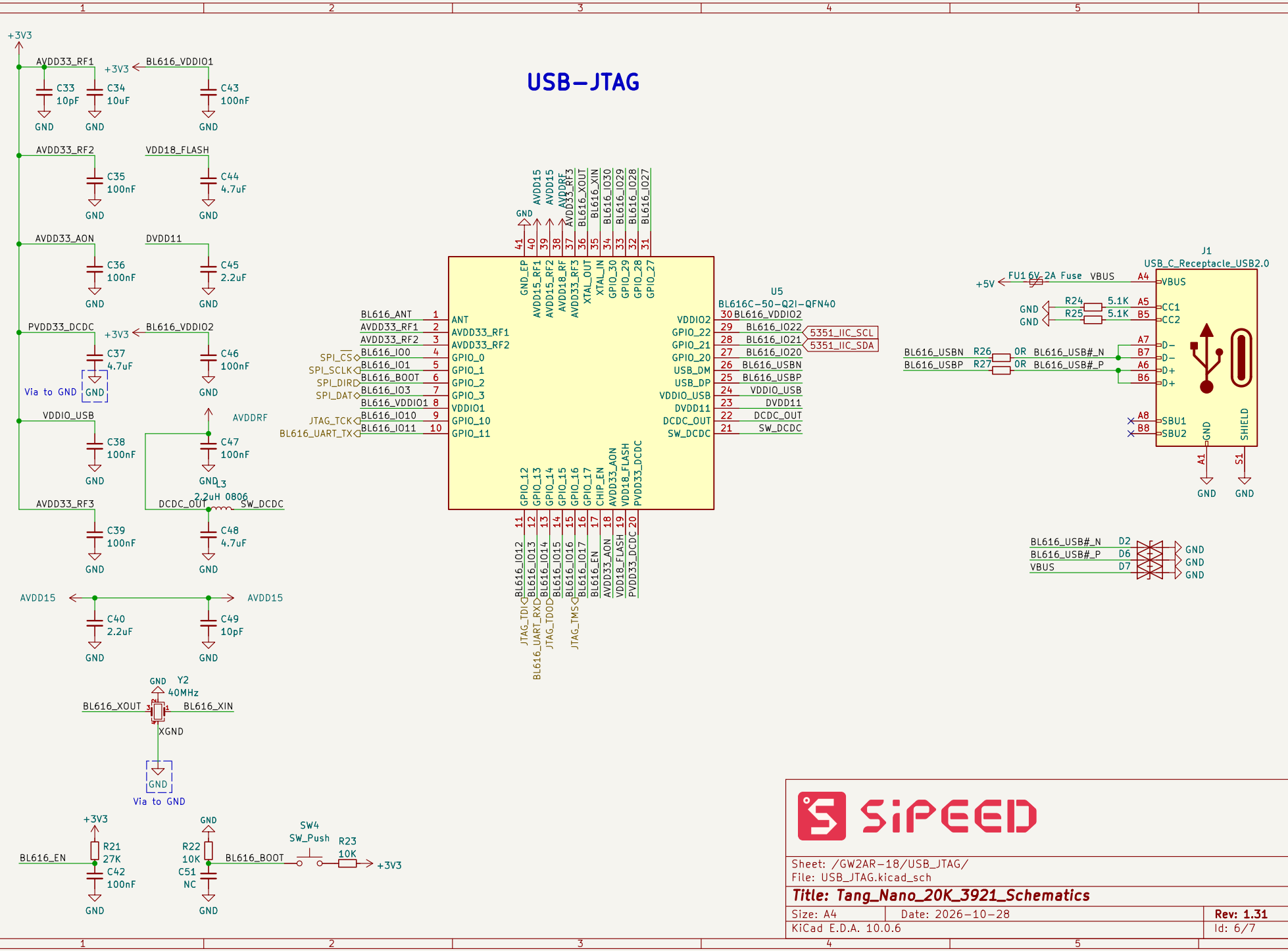


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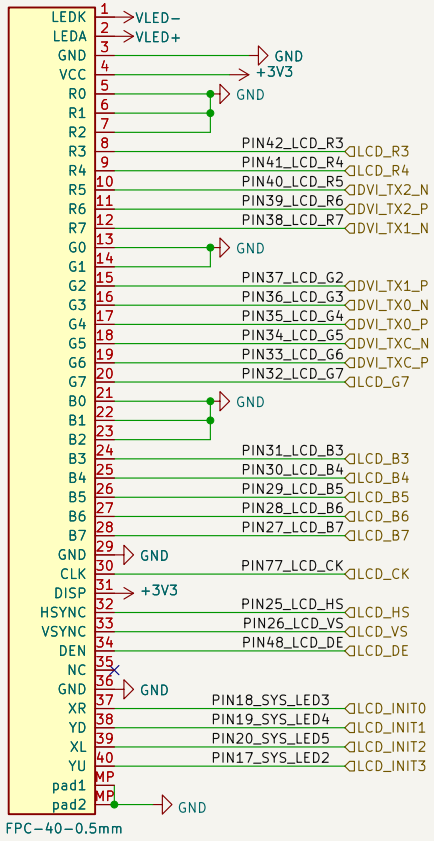
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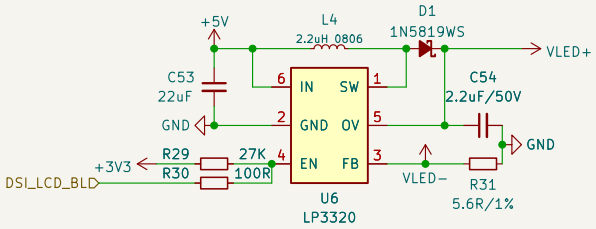
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USB-JTAG
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BL616C-50-Q2I-QFN40
J1
USB_C_Receptacle_USB2.0
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Date: 2026-10-28
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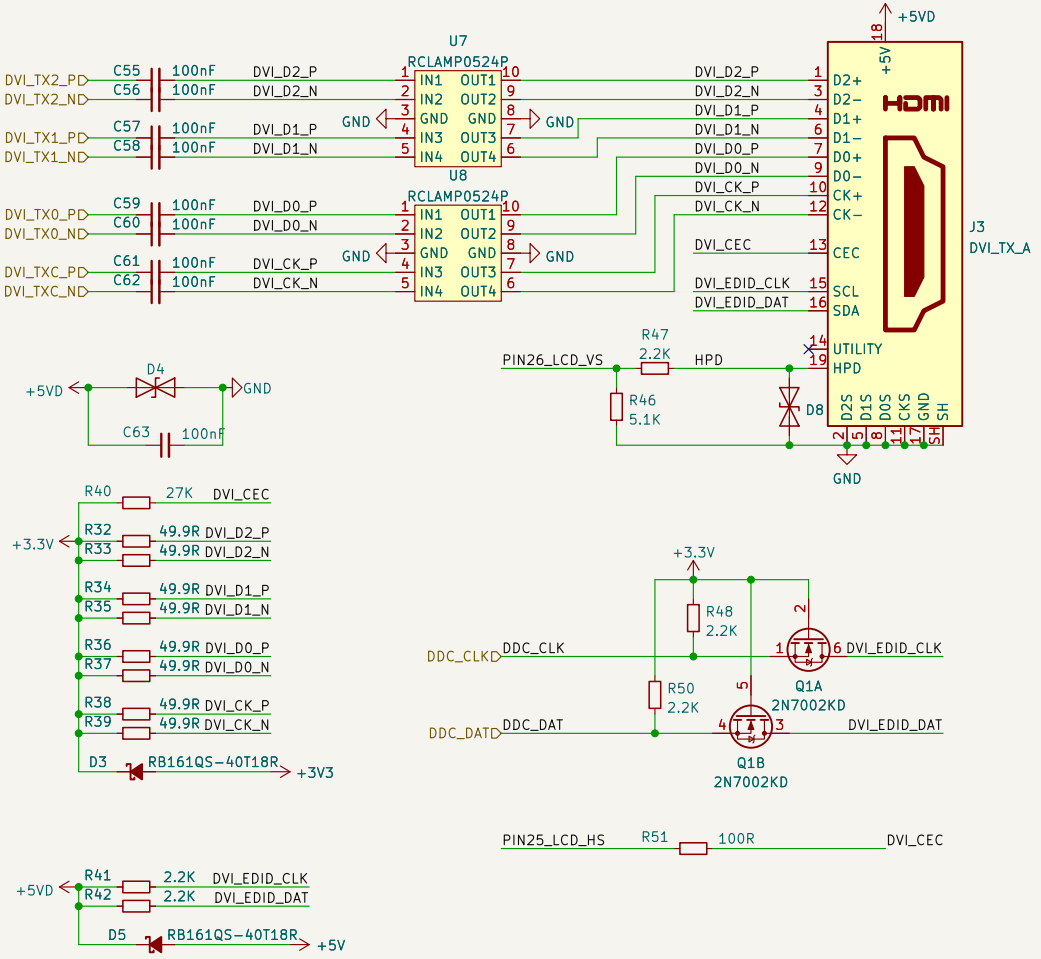
RGB LCD



LCD BACKLIGHT



DVI TX



SiPEED

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Title: Tang_Nano_20K_3921_Schematics

Size: A4	Date: 2026−10−28	**Rev: 1.31**
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